

Nash's theorem via Günther's trick

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Abstract

We present Günther's approach to Nash's embedding theorem with an emphasis on accessibility and rigor.

Riemannian geometry has two fundamental theorems. One is about the existence and uniqueness of the Levi-Civita connection, and the other is a result of John Nash [11], which states that *any Riemannian manifold is isometric to a smooth submanifold of Euclidean space*. In other words, Riemannian manifolds can be defined as submanifolds of Euclidean space.

The original proof is based on a delicate and, at the time, unprecedented iteration scheme with smoothing operators designed to overcome the loss of derivatives. This construction led to the Nash–Moser inverse function theorem, a method that has found many applications.

Later, Matthias Günther discovered a magic trick [5] that removes this loss-of-derivatives problem. His proof is based on the standard inverse function theorem in Banach spaces, which is, in essence, a contraction argument.

Thanks to Günther's simplification, I was able to include this theorem in a differential geometry course, and I hope it is worth sharing. Except for the inverse function theorem, I mostly follow Nash's original proof; I also use several cosmetic tweaks by Deane Yang [16], Terence Tao [14], and Ralph Howard [7]. Günther's approach is also carefully covered in two textbooks: by Michael Taylor [15] and by Qing Han and Jia-Xing Hong [6].

The reader is assumed to be familiar with smooth manifolds (charts, atlases, coverings, partitions of unity, and tensor fields). I use other results, including Schauder's estimates and the Whitney embedding theorem, but they may be treated as black boxes.

We formulate the embedding problem in Section 1. Section 2 introduces Nash's twist and solves an approximate version of the embedding problem. Section 3 consists of several steps that reduce the

main theorem to a perturbation problem of a metric on the torus. In Section 4, we introduce free maps and formulate Nash's lemma and Günther's lemma. In the following section we use these lemmas to solve the perturbation problem and therefore the embedding problem. The final two sections prove the lemmas. Günther's trick is explained in the last section; the rest of the paper is needed to appreciate it.

Here is a list of globally fixed notation.

$v \oplus w$, 3	dw, w^*, xw , 2	n, N , 7
$v \ominus w$, 6	Δu , 15	Q, Q_{ij} , 10
$dx_i \odot dx_j$, 5	L_{ij} , 14	$\tilde{Q}$, 11
$(d\psi)^2$, 4	L, M , 11	$T, T^\perp, T^2, N$, 13

1 Formulation

Induced metric. Given a smooth n -dimensional manifold Ω , we denote by $T\Omega$ and $T_p\Omega$ its tangent bundle and the tangent space at a point $p \in \Omega$, respectively.

Recall that a metric tensor, say g , on Ω is a smooth field of symmetric bilinear forms on $T\Omega$. Here and further, smooth means C^∞ .

Let $\bar{\Omega}$ be another smooth manifold, and let $w: \Omega \rightarrow \bar{\Omega}$ be a smooth map. Recall that the derivative of w in the direction of a tangent vector $x \in T_p\Omega$ is denoted by xw , and $xw = dw(x)$; that is, the derivative of w in the direction x is the differential $dw: T\Omega \rightarrow T\bar{\Omega}$ of w , evaluated at x . Note that $xw \in T_{w(p)}\bar{\Omega}$.

Suppose that $\bar{\Omega}$ is equipped with a metric tensor $\bar{g}$. Then the metric tensor g on Ω defined by

$$g(x, y) = \bar{g}(xw, yw)$$

is called the induced metric tensor, or the pullback of $\bar{g}$ to Ω . This relation can also be written as $g = w^*\bar{g}$. Equivalently, we may say that $w: (\Omega, g) \rightarrow (\bar{\Omega}, \bar{g})$ is isometric.

A metric tensor g is called Riemannian if it is *positive-definite*; that is, $g(x, x) > 0$ for any nonzero tangent vector x . A smooth manifold equipped with a Riemannian metric is called a Riemannian manifold.

Notice that if $g = w^*\bar{g}$ is a Riemannian metric, then w must be regular; that is, the differential $d_p w$ has rank n at every point $p \in \Omega$. In particular, w is an immersion.

We consider the space $\mathbb{R}^d$ equipped with the metric tensor defined by the standard scalar product:

$$\bar{g}(x, y) := \langle x, y \rangle = x_1 \cdot y_1 + \cdots + x_d \cdot y_d,$$

where x_i and y_i denote the coordinates of vectors $X, Y \in \mathbb{R}^d$. Note that $\bar{g}$ is Riemannian.

1.1. Main theorem. *Any n -dimensional Riemannian manifold (Ω, g) admits a smooth isometric embedding into a Euclidean space $\mathbb{R}^d$ for some d . Moreover, d can be bounded in terms of n .*

We give a complete proof only for compact manifolds and indicate the reduction of the theorem to the compact case; see Step 4 in the next section.

Remark on d . I do not make any effort to optimize the bound on d in terms of n . Tracing the estimates in the argument yields that $d = 10 \cdot n^4$ suffices.

For compact surfaces one can take $d = 5$; this seems to be the only nontrivial case when the optimal dimension is known. The existence is proved in [3, 3.2.4(D)], and there is no isometric embedding of $\mathbb{RP}^2$ with the standard metric into $\mathbb{R}^4$ [4, Appendix 4].

The bound $d \geq \binom{n+1}{2}$ is proved in [4, Appendix 1]. It means that the system

$$\textcircled{1} \quad \langle \partial_i w, \partial_j w \rangle = g_{ij},$$

cannot be overdetermined. This might be the optimal bound for large n and for the local version of the theorem; that is, *any point in a smooth n -dimensional Riemannian manifold has a neighborhood U that admits an isometric embedding into $\mathbb{R}^d$.*

The system $\textcircled{1}$ makes perfect sense for C^1 maps w , but the proof of the bound $d \geq \binom{n+1}{2}$ essentially uses that w is C^m -smooth for sufficiently large m . Without this assumption, the statement does not hold. Indeed, the Nash–Kuiper theorem [8, 10] states that $\textcircled{1}$ admits a C^1 solution w if $d = n + 1$. For instance, it might happen that the optimal dimension for C^2 -embeddings is smaller than $\binom{n+1}{2}$. This question was mentioned by Mikhael Gromov and Vladimir Rokhlin in the cited paper. Counting degrees of freedom of the curvature tensor gives a lower bound that is still quadratic in n ; namely, $d \geq \frac{1}{6} \cdot n \cdot (n+5)$ (check also my question on MathOverflow [13]).

2 Approximate theorem

Sum and direct sum. Consider two smooth maps $v_1: \Omega \rightarrow \mathbb{R}^{d_1}$ and $v_2: \Omega \rightarrow \mathbb{R}^{d_2}$. The map $v_1 \oplus v_2: p \mapsto (v_1(p), v_2(p))$ will be called the direct sum of v_1 and v_2 ; it maps Ω to $\mathbb{R}^{d_1+d_2} = \mathbb{R}^{d_1} \oplus \mathbb{R}^{d_2}$.

Suppose that metric tensors g , g_1 , and g_2 are induced by $v_1 \oplus v_2$, v_1 , and v_2 , respectively. Then

$$g = g_1 + g_2.$$

Furthermore, if $d_1 = d_2$, then we can consider the usual sum $v_1 + v_2$. Recall that the support of a map is the closure of the set where it takes nonzero values. Now suppose that g , g_1 , and g_2 are induced by $v_1 + v_2$, v_1 and v_2 , and the maps v_1 and v_2 have disjoint supports. Then again we have

$$g = g_1 + g_2.$$

These two observations will be used repeatedly without notice.

Nash's twist. Let φ and ψ be smooth functions on a smooth manifold Ω . Given $r > 0$, denote by $\mathbb{S}_r^1$ the circle of radius r in $\mathbb{R}^2$.

Nash's twist $\Theta: \Omega \rightarrow \mathbb{R}^2$ of the triple (r, φ, ψ) is defined as the composition

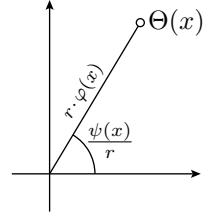
$$\Omega \xrightarrow{\psi} \mathbb{R} \xrightarrow{\ell_r} \mathbb{S}_r^1 \xrightarrow{\times \varphi} \mathbb{R}^2,$$

where $\ell_r: \mathbb{R} \rightarrow \mathbb{S}_r^1$ is a length-preserving covering map, say

$$\ell_r(x) = (r \cdot \cos \frac{x}{r}, r \cdot \sin \frac{x}{r}),$$

and $\times \varphi$ denotes multiplication by φ ; so

$$\Theta(x) := \varphi(x) \cdot (\ell_r \circ \psi(x)).$$



2.1. Claim. Nash's twist Θ for the triple (r, φ, ψ) induces the metric

$$g = \varphi^2 \cdot (d\psi)^2 + r^2 \cdot (d\varphi)^2.$$

Here $(d\psi)^2$ denotes the metric tensor induced by $\psi: \Omega \rightarrow \mathbb{R}$; that is,

$$(d\psi)^2(X, Y) := d\psi(X) \cdot d\psi(Y) = (X\psi) \cdot (Y\psi).$$

Computations. Suppose that X is a tangent vector on Ω . Then,

$$\begin{aligned} X\Theta &= X(\varphi \cdot \ell_r \circ \psi) = \\ &= (X\varphi) \cdot (\ell_r \circ \psi) + \varphi \cdot (\ell'_r \circ \psi) \cdot (X\psi). \end{aligned}$$

Observe that $|\ell_r| = r$, $|\ell'_r| = 1$, and $\ell_r \perp \ell'_r$. Therefore,

$$g(X, Y) = \langle X\Theta, Y\Theta \rangle = r^2 \cdot (X\varphi) \cdot (Y\varphi) + \varphi^2 \cdot (X\psi) \cdot (Y\psi). \quad \square$$

Approximate version. The following statement can be regarded as an approximate version of the main theorem (1.1). It implies in particular that the set of metrics induced by smooth maps into Euclidean space is dense in the space of all Riemannian metrics.

2.2. Proposition. *Let g be a Riemannian metric on a smooth n -manifold Ω . Then there exists a metric tensor h on Ω and a one-parameter family of smooth maps $w_t: \Omega \rightarrow \mathbb{R}^d$ with the induced metrics $g_t = g + t \cdot h$ for $t > 0$.*

The proof will produce a nonnegative metric tensor h , but this is not going to be used.

Observe that $g_t \rightarrow g$ as $t \rightarrow 0$. Therefore, one might be tempted to take the limit of w_t as $t \rightarrow 0$. However, as we will see, the maps w_t constructed in the proof converge to a constant; thus, the limit does not solve the problem.

Let $w: \Omega \rightarrow \mathbb{R}^d$ be a smooth map. Suppose that $w_1, \dots, w_d$ are the coordinate functions of w . Then the induced metric tensor g can be written as

$$\textbf{1} \quad g = (dw_1)^2 + \dots + (dw_d)^2.$$

In the main theorem, we need to find functions $w_1, \dots, w_d: \Omega \rightarrow \mathbb{R}$ that satisfy equation **1** for a given Riemannian metric g . The following weaker form of **1** will play a key role in the proof of 2.2.

2.3. Proposition. *Let (Ω, g) be a compact n -dimensional Riemannian manifold. Then there are smooth functions $\varphi_1, \dots, \varphi_d, \psi_1, \dots, \psi_d$ on Ω such that*

$$\textbf{2} \quad g = \varphi_1^2 \cdot (d\psi_1)^2 + \dots + \varphi_d^2 \cdot (d\psi_d)^2.$$

Moreover, d depends only on n .

Proof. The metric tensor g can be written in local coordinates $(x_1, \dots, x_n)$ as

$$\textbf{3} \quad g = \sum_{i,j} g_{ij} \cdot dx_i \odot dx_j,$$

where $g_{ij} = g_{ji}$ are smooth functions of $(x_1, \dots, x_n)$ and $\odot$ denotes the symmetric tensor product.

Since g is Riemannian, at any point $p \in \Omega$, we can choose a chart so that the vectors ∂_i are orthonormal at p ; that is, $g_{ii} = 1$ and $g_{ij} = 0$ at p for all $i \neq j$. Since g_{ij} are smooth functions, for any $\varepsilon > 0$, we can find a neighborhood $U \ni p$ such that

$$\textbf{4} \quad g_{ii} \leq 1 \pm \varepsilon \quad \text{and} \quad g_{ij} \leq \pm \varepsilon$$

for all $i \neq j$ and any point in U .

Observe that

$$\pm dx_i \odot dx_j = \frac{1}{2} \cdot (dx_i \pm dx_j)^2 - \frac{1}{2} \cdot (dx_i)^2 - \frac{1}{2} \cdot (dx_j)^2.$$

Let us plug this formula in ❸ and take ❹ into account. Assuming that ε is small, we will get that g is a linear combination with positive coefficients of the metric tensors $(dx_i)^2$ and $(dx_i \pm dx_j)^2$ for all i and j . In other words, we can take the functions x_i and $x_i \pm x_j$ for all $i < j$ as our functions $\psi_1, \dots, \psi_d$ (these should be extended smoothly to the whole manifold) and find $\varphi_1, \dots, \varphi_d$ such that ❷ holds in a neighborhood of p . Applying a partition of unity, we get the global statement.

It suffices to take n^2 functions in each chart. So, if Ω is covered by m charts, then we may take $d = n^2 \cdot m$. Moreover, the same bound applies if the charts can be colored using m colors such that distinct charts of the same color do not overlap. By the colored version of Lebesgue covering dimension [12, Theorem 2], we can assume that $m = n + 1$; therefore, $d = n^2 \cdot (n + 1)$ suffices. $\square$

Proof of 2.2. Suppose that $\varphi_1, \dots, \varphi_d, \psi_1, \dots, \psi_d$ are provided by 2.3. Let $r = \sqrt{t}$; denote by Θ_i Nash's twist for the triple (r, φ_i, ψ_i) . By 2.1,

$$w_t = \Theta_1 \oplus \dots \oplus \Theta_d$$

is the needed map with $h = (d\varphi_1)^2 + \dots + (d\varphi_d)^2$. $\square$

A pseudoeuclidean digression. Let us denote by $\mathbb{R}^r \ominus \mathbb{R}^s$ the pseudoeuclidean space with signature (r, s) ; that is, the space $\mathbb{R}^{r+s}$ with the scalar product defined by

$$\langle X, Y \rangle = x_1 \cdot y_1 + \dots + x_r \cdot y_r - x_{r+1} \cdot y_{r+1} - \dots - x_{r+s} \cdot y_{r+s},$$

where x_i and y_i denote the coordinates of vectors X and Y in $\mathbb{R}^{r+s}$.

2.4. Problem. Show that any metric g on a smooth manifold Ω is induced by a smooth embedding $\Omega \hookrightarrow \mathbb{R}^r \ominus \mathbb{R}^s$ for some large r and s .

Solution based on Nash's theorem. Notice that any metric g can be written as the difference $g = g_1 - g_2$ of two Riemannian metrics g_1 and g_2 on Ω . By Nash's theorem, we can find two smooth embeddings $w_1, w_2: \Omega \hookrightarrow \mathbb{R}^d$ whose induced metrics are g_1 and g_2 , respectively. It remains to observe that g is induced by

$$w_1 \ominus w_2: p \mapsto (w_1(p), w_2(p)) \in \mathbb{R}^d \ominus \mathbb{R}^d. \quad \square$$

The above solution uses the main theorem, which we have not yet proved. The following solution is more direct and relies only on Nash's twist.

Direct solution. Assume g is Riemannian, and let $\varphi_1, \dots, \varphi_d, \psi_1, \dots, \psi_d$ be provided by 2.3. Let Θ_i be Nash's twist for the triple $(1, \varphi_i, \psi_i)$, and

$$\begin{aligned} h &= (d\varphi_1)^2 + \dots + (d\varphi_d)^2, \\ \Theta &= \Theta_1 \oplus \dots \oplus \Theta_d: \Omega \rightarrow \mathbb{R}^{2 \cdot d}, \\ \varphi &= \varphi_1 \oplus \dots \oplus \varphi_d: \Omega \rightarrow \mathbb{R}^d. \end{aligned}$$

According to 2.1, the map Θ induces $g + h$. Note that φ induces h . Therefore $w = \Theta \ominus \varphi: \Omega \rightarrow \mathbb{R}^{2 \cdot d} \ominus \mathbb{R}^d$ induces g .

To handle the general case, express the metric as a difference of two Riemannian metrics: $g = g_1 - g_2$. As shown above, there exist smooth embeddings w_1 and w_2 into pseudoeuclidean spaces inducing g_1 and g_2 , respectively. It follows that $w_1 \ominus w_2$ induces g .

Finally, if $w_3: \Omega \hookrightarrow \mathbb{R}^{2 \cdot n+1}$ is an embedding provided by the Whitney embedding theorem, then $(w_1 \oplus w_3) \ominus (w_2 \oplus w_3)$ is the required smooth isometric embedding. $\square$

3 Reductions

Torus. Let us denote by $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ the n -dimensional torus.

The torus is a smooth manifold; it comes with global cyclic coordinates, which can be used to define the partial derivatives $\partial_1, \dots, \partial_n$. A metric tensor on $\mathbb{T}^n$ can be written in these coordinates; therefore, we can (and will) regard a metric tensor as a smooth map $g: \mathbb{T}^n \rightarrow \mathbb{R}^N$, where

$$N := \frac{n(n+1)}{2}.$$

Perturbation. The next theorem will be proved in the following sections; note that this is a very special case of the main theorem.

3.1. Theorem. *There exists a Riemannian metric g_0 on $\mathbb{T}^n$ such that any metric g that is sufficiently close to g_0 in the C^∞ topology is induced by a smooth map $\mathbb{T}^n \rightarrow \mathbb{R}^d$ for some d depending only on n .*

Now we will reduce the main theorem (1.1) to this restricted version. The reduction proceeds in several steps.

Step 1. Let us get an isometric immersion for any metric on the torus; namely,

❶ *given a Riemannian metric g on $\mathbb{T}^n$, there exists an isometric immersion $(\mathbb{T}^n, g) \looparrowright \mathbb{R}^d$ for some d depending only on n .*

Indeed, let g_0 be the metric from 3.1. After rescaling if necessary, we may assume that $g_0 < g$; that is, $g_0(X, X) < g(X, X)$ for any nonzero tangent vector X . In particular, the difference $g_1 = g - g_0$ is a Riemannian metric.

By 2.2, there exists a metric tensor h such that for any $t > 0$, there is a smooth map $w_1: \mathbb{T}^n \rightarrow \mathbb{R}^{d_1}$ that induces $g_1 + t \cdot h$.

By 3.1, for any small $t > 0$, there is a smooth map $w_0: \mathbb{T}^n \rightarrow \mathbb{R}^{d_0}$ that induces the metric $g_0 - t \cdot h$. Therefore, $w_0 \oplus w_1$ induces

$$g = g_0 - t \cdot h + g_1 + t \cdot h.$$

Step 2. Now let us improve the statement to an embedding; namely,

❷ *given a Riemannian metric g on $\mathbb{T}^n$, there is an isometric embedding $(\mathbb{T}^n, g) \hookrightarrow \mathbb{R}^d$ for some d depending only on n .*

Choose your favorite smooth regular embedding $w_1: \mathbb{T}^n \hookrightarrow \mathbb{R}^{n+1}$. Let g_1 be the metric induced by w_1 . Since w_1 is an regular, g_1 must be Riemannian.

We can assume that $g > g_1$; in other words the metric $g_2 = g - g_1$ is Riemannian. If this is not the case, replace w_1 with its rescaling $\varepsilon \cdot w_1$ for sufficiently small $\varepsilon > 0$.

Step 1 provides an isometric immersion $w_2: (\mathbb{T}^n, g_2) \looparrowright \mathbb{R}^d$. Notice that

$$w = w_1 \oplus w_2: (\mathbb{T}^n, g = g_1 + g_2) \rightarrow \mathbb{R}^{d+n+1}$$

is an isometric embedding. Indeed, w is isometric, and it is an embedding since w_1 is.

Step 3. Now we extend the statement to all compact manifolds; namely,

❸ *any compact n -dimensional Riemannian manifold (Ω, g) admits a smooth isometric embedding into $\mathbb{R}^d$ for some d that depends only on n .*

Since Ω is compact, the Whitney embedding theorem provides a smooth embedding $\iota: \Omega \hookrightarrow \mathbb{T}^{2n+1}$.

The Riemannian metric on Ω can be extended to a Riemannian metric on the torus. This follows, for example, by choosing a metric on a tubular neighborhood that restricts to the given one on Ω ,

and then patching it with any background metric on the torus via a partition of unity. It remains to take a smooth isometric embedding $w: (\mathbb{T}^{2 \cdot n+1}, \bar{g}) \hookrightarrow \mathbb{R}^d$ provided by Step 2 and observe that the composition $w \circ \iota: (\Omega, g) \hookrightarrow \mathbb{R}^d$ is an isometric embedding.

Step 4. Now we sketch reduction of the main theorem to the compact case provided in Step 3. A complete proof is given in [11, Part D].

Applying an argument similar to Step 2, we can reduce the question to finding an isometric immersion of (Ω, g) .

Further, we need to find a sequence of maps $r_i: \Omega \rightarrow \mathbb{S}^n$ and a sequence of Riemannian metrics h_i on $\mathbb{S}^n$ such that r_i is constant outside of a compact subset $K_i \subset \Omega$ and

$$g = \sum_i r_i^* h_i.$$

This part is done by applying a partition of unity.

In addition, we have to arrange this construction so that the index set $\mathcal{I}$ can be colored into $n+1$ colors such that $K_i \cap K_j = \emptyset$ for distinct indices i and j of the same color. More precisely, $\mathcal{I}$ can be partitioned into $n+1$ subsets $\mathcal{I}_0, \dots, \mathcal{I}_n$ such that for any k and any distinct $i, j \in \mathcal{I}_k$, we have $K_i \cap K_j = \emptyset$. This is very similar to the colored version of Lebesgue covering dimension.

Applying the compact case of the theorem to each $(\mathbb{S}^n, h_i)$, we get a sequence of isometric embeddings $w_i: (\mathbb{S}^n, h_i) \hookrightarrow \mathbb{R}^d$. By shifting w_i , we can assume that $w_i \circ r_i$ has support in K_i . Now note that the map

$$\left(\sum_{i \in \mathcal{I}_0} w_i \circ r_i \right) \oplus \dots \oplus \left(\sum_{i \in \mathcal{I}_n} w_i \circ r_i \right)$$

gives the required isometric immersion $(\Omega, g) \hookrightarrow \mathbb{R}^{(n+1) \cdot d}$.

4 Free maps and key lemmas

Here we introduce free maps and formulate two key lemmas that will be used in the next section.

From now on, we look only at the torus case. Recall that our torus $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ is equipped with cyclic coordinates and globally defined partial derivatives $\partial_1, \dots, \partial_n$. Furthermore, a metric tensor on $\mathbb{T}^n$ can be described by a smooth map $g: \mathbb{T}^n \rightarrow \mathbb{R}^N$ with components g_{ij} for $1 \leq i \leq j \leq n$; here

$$N := \frac{n \cdot (n+1)}{2}.$$

Free Maps. Recall that a smooth map $f: \mathbb{T}^n \rightarrow \mathbb{R}^d$ is regular if its differential df has rank n at every point. In other words, for any point $p \in \mathbb{T}^n$, the first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ are linearly independent.

A map $f: \mathbb{T}^n \rightarrow \mathbb{R}^d$ is called free (or two-regular) if an analogous property holds for both first- and second-order partial derivatives: that is, at any point $p \in \mathbb{T}^n$, all $n + \frac{n(n+1)}{2}$ partial derivatives $\partial_i f, \partial_{ij} f$ for $i \leq j$ are linearly independent.

Let $\psi_\ell: \mathbb{T}^n \rightarrow \mathbb{S}^1$ be a map induced by a linear function $\ell: \mathbb{R}^n \rightarrow \mathbb{R}$. Let us regard $\mathbb{S}^1$ as a subset of $\mathbb{R}^2$ and consider the map $f: \mathbb{T}^n \rightarrow \mathbb{R}^{n \cdot (n+1)}$ defined as the direct sum of the maps ψ_{x_i} and $\psi_{x_i + x_j}$, where x_i are the coordinate functions on $\mathbb{R}^n$ and $i < j$. It is straightforward to check that f is free, so we have proved the following claim.

4.1. Claim. *There is a smooth free embedding $f: \mathbb{T}^n \hookrightarrow \mathbb{R}^{n \cdot (n+1)}$.*

Q-form. Now we introduce a fancy notation for the induced metric. We define a symmetric bilinear form Q that takes two smooth maps $v, w: \mathbb{T}^n \rightarrow \mathbb{R}^d$ and returns the following metric tensor on $\mathbb{T}^n$:

$$g(X, Y) := \frac{1}{2} \cdot [\langle Xv, Yw \rangle + \langle Xw, Yv \rangle].$$

Recall that the cyclic coordinates on $\mathbb{T}^n$ allow us to write a metric tensor as a smooth map $\mathbb{T}^n \rightarrow \mathbb{R}^N$. The components of the metric tensor can be written as

$$g_{ij} = Q_{ij}(v, w) = \frac{1}{2} \cdot [\langle \partial_i v, \partial_j w \rangle + \langle \partial_i w, \partial_j v \rangle].$$

Notice that a metric tensor g is induced by a smooth map $w: \mathbb{T}^n \rightarrow \mathbb{R}^d$ if and only if $g = Q(w, w)$.

Hölder norms. Fix $\alpha \in (0, 1)$, say $\alpha = \frac{1}{2}$. Denote by $|x - y|_{\mathbb{T}^n}$ the standard distance between points $x, y \in \mathbb{T}^n$. Let us recall the definition of the Hölder norms on an open subset $W \subset \mathbb{T}^n$

$$\begin{aligned} \|u\|_{C^{0,\alpha}(W)} &:= \sup_x \{ |u(x)| \} + \sup_{x \neq y} \left\{ \frac{|u(x) - u(y)|}{|x - y|_{\mathbb{T}^n}^\alpha} \right\}, \\ \|u\|_{C^{k,\alpha}(W)} &:= \max_{|\mathcal{I}| \leq k} \left\{ \|\partial^{\mathcal{I}} u\|_{C^{0,\alpha}(W)} \right\}, \end{aligned}$$

where $x, y \in W$, and $\mathcal{I} = (i_1, \dots, i_n)$ denotes a multi-index; so, $\partial^{\mathcal{I}}$ is the partial derivative $\partial_1^{i_1} \dots \partial_n^{i_n}$ of order $|\mathcal{I}| := i_1 + \dots + i_n$. The space of functions $W \rightarrow \mathbb{R}$ with finite $C^{k,\alpha}(W)$ -norm is called the (k, α) -Hölder space on W and is denoted by $C^{k,\alpha}(W)$; we may suppress W in this notation.

Let us define $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R}^d)$ as the space of functions $u: \mathbb{T}^n \rightarrow \mathbb{R}^d$ whose coordinate functions $u_1, \dots, u_d$ belong to $C^{k,\alpha}(\mathbb{T}^n)$, and set

$$\|u\|_{C^{k,\alpha}} := \max\{\|u_1\|_{C^{k,\alpha}}, \dots, \|u_d\|_{C^{k,\alpha}}\}.$$

Key lemmas. From now on, let us fix a smooth free embedding $f: \mathbb{T}^n \hookrightarrow \mathbb{R}^d$ (which exists by 4.1) and define the linear operator

$$L: C^\infty(\mathbb{T}^n, \mathbb{R}^d) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^N) \quad \text{by} \quad L(w) := 2 \cdot Q(f, w).$$

4.2. Nash's lemma. *The linear operator L admits a right inverse*

$$M: C^\infty(\mathbb{T}^n, \mathbb{R}^N) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^d)$$

such that

$$\|Mh\|_{C^{2,\alpha}} \leq a \cdot \|h\|_{C^{2,\alpha}}$$

for some constant a and any $h \in C^\infty(\mathbb{T}^n, \mathbb{R}^N)$.

4.3. Günther's lemma. *There exists a symmetric bilinear form*

$$\tilde{Q}: C^\infty(\mathbb{T}^n, \mathbb{R}^d) \times C^\infty(\mathbb{T}^n, \mathbb{R}^d) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^d)$$

and a sequence of constants $b_2, b_3, \dots$ such that $L\tilde{Q} = Q$ and

$$\|\tilde{Q}(w, w)\|_{C^{2,\alpha}} \leq b_2 \cdot \|w\|_{C^{2,\alpha}}^2,$$

$$\|\tilde{Q}(w, w)\|_{C^{k,\alpha}} \leq b_2 \cdot \|w\|_{C^{2,\alpha}} \cdot \|w\|_{C^{k,\alpha}} + b_k \cdot \|w\|_{C^{k-1,\alpha}}^2$$

for any $w \in C^\infty(\mathbb{T}^n, \mathbb{R}^d)$.

Remarks. As we will see soon, the perturbation theorem follows if the quadratic equation $L(w) + Q(w, w) = h$ has a solution for any sufficiently small metric tensor h .

Note that Nash's lemma provides a solution of the corresponding linear equation $L(w) = h$.

There are many symmetric bilinear forms that solve the equation $L\tilde{Q} = Q$. In particular, we could take $\tilde{Q} = MQ$, where M is provided by Nash's lemma. However, this choice of $\tilde{Q}$ does not satisfy the inequalities required by Günther's lemma. These inequalities are essential; they enable the use of the contraction mapping theorem in the following proof.

5 Perturbation

We now complete the proof of Nash's theorem, modulo the two lemmas from the previous section. Recall that $f: \mathbb{T}^n \hookrightarrow \mathbb{R}^d$ is a fixed smooth free embedding and $L(w) := 2 \cdot Q(f, w)$.

5.1. Lemma. *There exists $r > 0$ such that the equation*

$$\textcircled{1} \quad L(w) + Q(w, w) = h$$

admits a smooth solution $w: \mathbb{T}^n \rightarrow \mathbb{R}^d$ for any smooth metric h (not necessarily Riemannian) with $\|h\|_{C^{2,\alpha}} < r$. Moreover, we can assume that

$$\|w\|_{C^{2,\alpha}} \leq c \cdot \|h\|_{C^{2,\alpha}}$$

for some constant c .

In the previous section, we showed that the perturbation theorem (3.1) implies the main theorem (1.1). Now we will show that the perturbation lemma implies the perturbation theorem (3.1), and therefore the main theorem (1.1).

Proof of 3.1 modulo 5.1. Let g be the metric tensor induced by f . Notice that the equation

$$Q(f + w, f + w) = g + h$$

is equivalent to $\textcircled{1}$. Indeed,

$$\begin{aligned} Q(f + w, f + w) &= Q(f, f) + 2 \cdot Q(f, w) + Q(w, w) \\ &= g + L(w) + Q(w, w). \end{aligned}$$

Hence, 3.1 follows. $\square$

In the proof of the perturbation lemma, we will follow the argument of the classical inverse function theorem in Banach spaces; see, for example, [9].

Proof of 5.1 modulo 4.2 and 4.3. Recall that M , $\tilde{Q}$, a , and b_2 are provided by 4.2 and 4.3. Set $R = \frac{1}{10 \cdot b_2}$ and $r = \frac{R}{10 \cdot a}$. Assume $\|h\|_{C^{2,\alpha}} \leq r$. By 4.3,

$$\Phi: w \mapsto Mh - \tilde{Q}(w, w)$$

is a contraction on the closed ball $\overline{B}[0, R] \subset C^{2,\alpha}(\mathbb{T}^n, \mathbb{R}^d)$.

Consider the sequence of maps $w_0, w_1, \dots: \mathbb{T}^n \rightarrow \mathbb{R}^d$ defined by $w_0 = 0$ and $w_{n+1} = \Phi(w_n)$ for all n . Notice that each w_n is smooth.

Since Φ is a contraction, the sequence w_n converges in $C^{2,\alpha}(\mathbb{T}^n, \mathbb{R}^d)$; denote its limit by w . Then

$$\|w\|_{C^{2,\alpha}} \leq R \quad \text{and} \quad w = Mh - \tilde{Q}(w, w).$$

Since $LM = \text{id}$ and $L\tilde{Q} = Q$, applying L to both sides yields that w solves ❶.

It remains to bootstrap regularity via the following claim:

❷ *The sequence $\|w_n\|_{C^{k,\alpha}}$ is bounded for every integer $k \geq 2$.*

Since the embedding $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R}^d) \hookrightarrow C^k(\mathbb{T}^n, \mathbb{R}^d)$ is compact, a subsequence of w_n converges in $C^k(\mathbb{T}^n, \mathbb{R}^d)$; but its limit must coincide with w . So w is C^k -smooth for every k and therefore smooth.

We prove ❷ by induction on k . The base case $k = 2$ has already been proved.

By the induction hypothesis, $\|w_n\|_{C^{k-1,\alpha}}$ is bounded. In particular, there is a constant c such that

$$\|Mh\|_{C^{k,\alpha}} + b_k \cdot \|w_n\|_{C^{k-1,\alpha}}^2 \leq c$$

for all n ; here b_k is the constant from 4.3.

Using the second estimate in 4.3, we get

$$\begin{aligned} \|w_{n+1}\|_{C^{k,\alpha}} &\leq \|Mh\|_{C^{k,\alpha}} + b_2 \cdot \|w_n\|_{C^{2,\alpha}} \cdot \|w_n\|_{C^{k,\alpha}} + b_k \cdot \|w_n\|_{C^{k-1,\alpha}}^2 \leq \\ &\leq c + b_2 \cdot R \cdot \|w_n\|_{C^{k,\alpha}} \leq \\ &\leq c + \frac{1}{10} \cdot \|w_n\|_{C^{k,\alpha}}. \end{aligned}$$

In particular,

$$\|w_n\|_{C^{k,\alpha}} \leq 2 \cdot c \quad \implies \quad \|w_{n+1}\|_{C^{k,\alpha}} \leq 2 \cdot c$$

for all n . Since $w_0 = 0$, we get ❷. □

6 Proof of Nash's lemma

Recall that $f: \mathbb{T}^n \hookrightarrow \mathbb{R}^d$ is a fixed smooth free embedding; in particular, f is regular.

The tangent space T_p is the n -dimensional space spanned by the first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ at p . Let $T_p^\perp$ denote the orthogonal complement of T_p in $\mathbb{R}^d$. Further, define the osculating space T_p^2 as the span of all first- and second-order partial derivatives $\partial_i f$ and $\partial_{ij} f$ at p , and set $N_p := T_p^2 \cap T_p^\perp$.

Since f is free, the dimensions of the spaces T_p , $T_p^\perp$, T_p^2 , and N_p are n , $d - n$, $n + \frac{n(n+1)}{2}$, and $\frac{n(n+1)}{2}$, respectively. We will write T , $T^\perp$, T^2 , and N for the corresponding vector bundles over $\mathbb{T}^n$.

Proof of 4.2. Recall that $\mathbb{T}^n$ has global cyclic coordinates.

Choose a metric tensor h on $\mathbb{T}^n$. Since all $\partial_i f$, $\partial_{ij} f$ are linearly independent, the equations $\langle \partial_{ij} f, w \rangle = -\frac{1}{2} \cdot h_{ij}$ for all i and j uniquely define an N -vector field w . Equivalently, w can be defined as a linear combination of $\partial_i f$, $\partial_{ij} f$ such that

$$\textcircled{1} \quad \langle \partial_i f, w \rangle = 0, \quad -2 \cdot \langle \partial_{ij} f, w \rangle = h_{ij}$$

at each point.

Notice that

$$\begin{aligned} L_{ij}(w) &= 2 \cdot Q_{ij}(f, w) = \\ &= \langle \partial_i f, \partial_j w \rangle + \langle \partial_j f, \partial_i w \rangle = \\ &= \partial_j \langle \partial_i f, w \rangle + \partial_i \langle \partial_j f, w \rangle - 2 \cdot \langle \partial_{ij} f, w \rangle. \end{aligned}$$

By $\textcircled{1}$, the operator $M: h \mapsto w$ serves as the desired right inverse. $\square$

The above argument yields the following.

6.1. Observation. *The right inverse M in Nash's lemma can be chosen so that Mh is an N -field; that is, $Mh(p) \in N_p$ for all $p \in \mathbb{T}^n$.*

Following the proof of Nash's lemma, we also get the following observation, which will play an essential role in the proof of Günther's lemma.

6.2. Observation. *Let $f: \mathbb{T}^n \rightarrow \mathbb{R}^d$ be a free map. Given a tangent vector field x on $\mathbb{T}^n$ and a metric tensor h on $\mathbb{T}^n$, there is a unique N -field w such that $L(x + w) = h$. In other words, given x , there is a right inverse $M_x: h \mapsto x + w$ of L , where w is an N -field.*

7 Proof of Günther's lemma

The form $\tilde{Q}$ can be defined by the formula

$$\tilde{Q}(w, w) = M_x Q(w, w),$$

where M_x is provided by 6.2 and x is a tangent vector field on $\mathbb{T}^n$ that depends on w ; see the remark right after the proof below. The trick is to choose x so that it absorbs high-frequency terms (see $\textcircled{3}$, $\textcircled{6}$, and $\textcircled{8}$ below) and dissolves the loss-of-derivatives problem.

The perturbations of f in 5.1 have essential tangential parts; this is the main difference from Nash's original construction. Note that perturbing a map in a tangential direction does not affect the Riemannian metric to first order; it only changes the parametrization. For this reason, adding to the construction the tangent field might

look strange. However, this idea works and similar ideas work in the DeTurck trick in the Ricci flow [1, 2.2] and Kuiper's improvement of Nash's C^1 -embedding theorem [8].

Smoothing operator. Recall that $\alpha \in (0, 1)$ is fixed, say $\alpha = \frac{1}{2}$.

Let $u: \mathbb{T}^n \rightarrow \mathbb{R}$ be a smooth function; its Laplacian is defined by

$$\Delta u = \sum_s \partial_{ss} u,$$

where $\partial_1, \dots, \partial_n$ are the standard partial derivatives on $\mathbb{T}^n$.

7.1. Proposition. *For any integer $k \geq 2$, the operator*

$$(\Delta - 1): u \mapsto \Delta u - u$$

defines a bi-Lipschitz isomorphism $C^{k,\alpha}(\mathbb{T}^n) \rightarrow C^{k-2,\alpha}(\mathbb{T}^n)$. In other words, $\Delta - 1$ is bounded and has bounded inverse $(\Delta - 1)^{-1}$.

Sketch. Let v be a smooth function on $\mathbb{T}^n$. Fourier analysis guarantees the existence of a unique smooth solution u to the equation $\Delta u - u = v$. Since smooth functions are dense in $C^{k-2,\alpha}(\mathbb{T}^n)$, it is sufficient to prove the following inequality for some constant c :

$$\textbf{①} \quad \frac{1}{c} \cdot \|v\|_{C^{k-2,\alpha}(\mathbb{T}^n)} \leq \|u\|_{C^{k,\alpha}(\mathbb{T}^n)} \leq c \cdot \|v\|_{C^{k-2,\alpha}(\mathbb{T}^n)}.$$

Note that the first inequality is trivial.

By the maximum principle, we have

$$\textbf{②} \quad \|u\|_{L^\infty(\mathbb{T}^n)} \leq \|v\|_{L^\infty(\mathbb{T}^n)},$$

where $\|\cdot\|_{L^\infty(W)}$ denotes the sup-norm on $W \subset \mathbb{T}^n$. Further, choose a $\frac{1}{10}$ -ball B in $\mathbb{T}^n$ (with the standard metric); denote by $2 \cdot B$ the concentric ball with twice the radius. Note that $2 \cdot B$ is isometric to a Euclidean ball, so Schauder's estimates can be applied to the restriction $u|_{2 \cdot B}$ (see for example [2, Lemma 6.1(a)]). This implies

$$\begin{aligned} \|u\|_{C^{2,\alpha}(B)} &\leq c \cdot (\|v\|_{C^{0,\alpha}(2 \cdot B)} + \|u\|_{L^\infty(2 \cdot B)}) \leq \\ &\leq c \cdot (\|v\|_{C^{0,\alpha}(\mathbb{T}^n)} + \|u\|_{L^\infty(\mathbb{T}^n)}). \end{aligned}$$

After averaging this inequality for balls in $\mathbb{T}^n$ and doing straightforward manipulations using **②**, we get the second inequality in **①**. $\square$

Notice that $(\Delta - 1)^{-1}$ is a smoothing operator; it takes a $C^{k-2,\alpha}$ -function and returns a $C^{k,\alpha}$ -function. Moreover, $(\Delta - 1)^{-1}$ commutes with partial derivatives; that is,

$$(\Delta - 1)^{-1} \partial_i = \partial_i (\Delta - 1)^{-1}.$$

These two properties will play a key role in the proof that follows.

In addition we will need the following claim that follows from the product rule for derivatives.

7.2. Claim. *Given a function $v \in C^\infty(\mathbb{T}^n)$, there is a sequence of constants $c_0, c_1, \dots$ such that*

$$\|u \cdot v\|_{C^{k,\alpha}} \leq c_0 \cdot \|u\|_{C^{k,\alpha}} + c_k \cdot \|u\|_{C^{k-1,\alpha}}$$

for any integer $k \geq 1$ and $u \in C^{k,\alpha}(\mathbb{T}^n)$.

Proof of 4.3. Recall that $Q_{ij}(w, w) = \langle \partial_i w, \partial_j w \rangle$. Let us show that

$$\textcircled{3} \quad (\Delta - 1)Q_{ij}(w, w) = A_{ij}(w) + \partial_i A_j(w) + \partial_j A_i(w),$$

where each A_{ij} , A_i and A_j is a linear combination (with constant coefficients) of the following quadratic forms

$$\textcircled{4} \quad w \mapsto \langle \partial_a w, \partial_b w \rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_c w \rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_{cd} w \rangle.$$

for all indices a, b, c, d .

Indeed,

$$\begin{aligned} (\Delta - 1)Q_{ij}(w, w) &= \langle \partial_i \Delta w, \partial_j w \rangle + \langle \partial_i w, \partial_j \Delta w \rangle + \\ &\quad + 2 \cdot \sum_s \langle \partial_{is} w, \partial_{js} w \rangle - \langle \partial_i w, \partial_j w \rangle = \\ &= 2 \cdot \sum_s \langle \partial_{is} w, \partial_{js} w \rangle - \langle \partial_i w, \partial_j w \rangle - 2 \cdot \langle \Delta w, \partial_{ij} w \rangle + \\ &\quad + \partial_i \langle \Delta w, \partial_j w \rangle + \partial_j \langle \Delta w, \partial_i w \rangle. \end{aligned}$$

This proves $\textcircled{3}$.

Now choose $A = A_i$ or $A = A_{ij}$ for some i and j . Notice that there is a sequence of constants $c_2, c_3, \dots$ such that

$$\begin{aligned} \textcircled{5} \quad &\|A(w)\|_{C^{0,\alpha}} \leq c_2 \cdot \|w\|_{C^{2,\alpha}}^2, \\ &\|A(w)\|_{C^{k-2,\alpha}} \leq c_2 \cdot \|w\|_{C^{k,\alpha}} \cdot \|w\|_{C^{2,\alpha}} + c_k \cdot \|w\|_{C^{k-1,\alpha}}^2. \end{aligned}$$

These inequalities follow by checking each form in $\textcircled{4}$. Let us do the case $w \mapsto \langle \partial_{ab} w, \partial_{cd} w \rangle$. The first inequality in $\textcircled{5}$ is trivial. The second follows since for any partial derivative $\partial^{\mathcal{I}}$ of order $k - 2$ we have

$$\partial^{\mathcal{I}} \langle \partial_{ab} w, \partial_{cd} w \rangle = \langle \partial_{ab} \partial^{\mathcal{I}} w, \partial_{cd} w \rangle + \langle \partial_{ab} w, \partial_{cd} \partial^{\mathcal{I}} w \rangle + \text{extra terms},$$

where each extra term has form $\langle \partial^{\mathcal{K}} w, \partial^{\mathcal{L}} w \rangle$ and partial derivatives $\partial^{\mathcal{K}}$ and $\partial^{\mathcal{L}}$ have order at most $k - 1$.

By 7.1, we have another sequence of constants $c_2, c_3, \dots$ such that

$$\begin{aligned} \textcircled{6} \quad & \|(\Delta - 1)^{-1}A(w)\|_{C^{2,\alpha}} \leq c_2 \cdot \|w\|_{C^{2,\alpha}}^2, \\ & \|(\Delta - 1)^{-1}A(w)\|_{C^{k,\alpha}} \leq c_2 \cdot \|w\|_{C^{k,\alpha}} \cdot \|w\|_{C^{2,\alpha}} + c_k \cdot \|w\|_{C^{k-1,\alpha}}^2. \end{aligned}$$

Recall also that

$$\textcircled{7} \quad L_{ij}(v) = -2 \cdot \langle \partial_{ij}f, v \rangle + \partial_i \langle \partial_j f, v \rangle + \partial_j \langle \partial_i f, v \rangle;$$

see the proof of Nash's lemma. Let us define $\tilde{Q}(w, w)$ as a linear combination of $\partial_i f$ and $\partial_{ij}f$ for all $i \leq j$ such that

$$\begin{aligned} \textcircled{8} \quad & -2 \cdot \langle \partial_{ij}f, \tilde{Q}(w, w) \rangle = (\Delta - 1)^{-1}A_{ij}(w), \\ & \langle \partial_i f, \tilde{Q}(w, w) \rangle = (\Delta - 1)^{-1}A_i(w) \end{aligned}$$

for all i and j . Since f is free, $\tilde{Q}(w, w)$ is well-defined, and it extends to a symmetric bilinear form. Combining $\textcircled{7}$, $\textcircled{8}$, and $\textcircled{3}$, we get

$$L\tilde{Q} = Q.$$

Let $\{e_i, e_{ij}\}_{i \leq j}$ be the dual frame to $\{\partial_i f, \partial_{ij}f\}_{i \leq j}$ in the osculating bundle T^2 . Notice that

$$\textcircled{9} \quad \tilde{Q}(w, w) = \sum_i (\Delta - 1)^{-1}A_i(w) \cdot e_i - \frac{1}{2} \cdot \sum_{i \leq j} (\Delta - 1)^{-1}A_{ij}(w) \cdot e_{ij}.$$

Since the maps $e_i, e_{ij}: \mathbb{T}^n \rightarrow \mathbb{R}^d$ are smooth, $\textcircled{6}$ together with 7.2 imply the inequalities in 4.3; in particular, the coefficient in front of $\|w\|_{C^{2,\alpha}} \cdot \|w\|_{C^{k,\alpha}}$ remains unchanged for all k . $\square$

The promised tangent vector field $x = x(w)$ is defined by the equations

$$\langle \partial_i f, x \rangle = (\Delta - 1)^{-1}A_i(w),$$

which is essentially the second line in $\textcircled{9}$. In other words, x is the first sum in the right-hand side of $\textcircled{9}$.

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